

5 Cyclin-dependent kinases (CDKs) are enzymes that regulate the cell cycle.

(a) Cell signalling by ligands causes the activation of CDKs in target cells.

Outline the main stages in the process of cell signalling by ligands that can cause specific responses in target cells.

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(b) The activity of CDKs is reduced by CDK inhibitors. Many of these inhibitors occur naturally in cells.

Fig. 5.1 is a diagram of a CDK inhibitor binding to a CDK molecule.

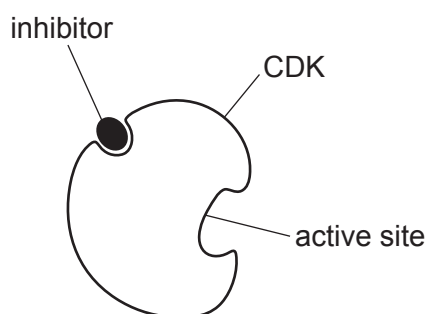


Fig. 5.1

State **and** explain how the CDK inhibitor in Fig. 5.1 prevents the activity of the CDK molecule.

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- (c) Table 5.1 lists three different CDKs, their roles in the cell cycle and molecules that inhibit them.

Table 5.1

name of CDK	role of CDK in the cell cycle	CDK inhibitor
CDK1	regulates cell progression from G ₂ to mitosis	RO-3306
CDK2	regulates the processes of the S phase	p21Cip1
CDK4	regulates cell progression from G ₁ to the S phase	palbociclib

- (i) With reference to Table 5.1, state which CDK inhibitor is likely to result in a cell containing **one** chromatid per chromosome. Explain your answer.

inhibitor

explanation

[2]

- (ii) With reference to Table 5.1, state which CDK inhibitor is likely to result in a cell with:

- a relatively high concentration of mitochondria
- **two** chromatids per chromosome.

Explain your answer.

inhibitor

explanation

[2]

- (d) Scientists have developed CDK inhibitors that are synthetic.

Explain why CDK inhibitors can be used to treat cancerous tumours.

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 [2]